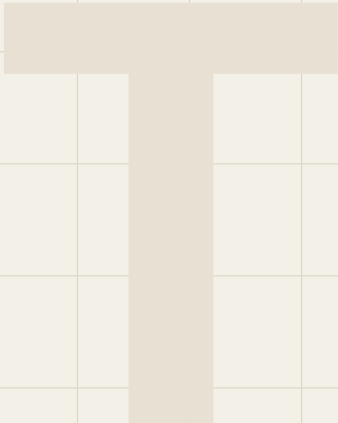


# TinyAI Protocol

## Sovereign On-chain AI State Machines for the EVM

把小型确定性模型变成可拥有、可验证、可进化的链上状态机。身份、记忆、大脑、组件与交易结算共享同一个公开执行环境。

|            |                  |
|------------|------------------|
| EXECUTION  | EVM NATIVE       |
| IDENTITY   | ERC-721          |
| EVOLUTION  | OWNER CONTROLLED |
| SETTLEMENT | NON-CUSTODIAL    |



# Contents

本白皮书只描述当前参考实现的协议机制、控制边界和独立验证方法，不以产品历史或营销叙事代替技术事实。

**中文**      中文执行摘要

**ABSTRACT**      Abstract

**01**      1. Protocol definition

**02**      2. Design principles

**03**      3. System architecture

**04**      4. AI identity and state

**05**      5. Deterministic inference

**06**      6. Conversation and memory

**07**      7. Brain publication and evolution

**08**      8. Component protocol

**09**      9. Market and settlement

**10**      10. Permission model

**11**      11. Economic surfaces

**12**      12. Security invariants

**13**      13. Independent verification

**14**      14. Composability

**15**      15. Research horizon

**16**      16. Conclusion

**A**      Appendix A. Core bounds

**B**      Appendix B. Terminology

# 中文执行摘要

TinyAI Protocol 是一套面向 EVM 的链上 AI 协议。它不把“连接钱包的中心化聊天机器人”定义为链上 AI，而是要求最终回答、模型路由、完整性校验和状态更新能够由区块链节点直接执行。

协议中的每只 AI 是一枚 ERC-721，也是一份独立的状态机。NFT 决定控制权，**AIState** 保存 DNA、记忆承诺、经验、性格、能力、大脑版本和升级策略。转移 NFT，就同时转移这只 AI 的完整链上身份和未来控制权。

大脑通过追加式注册表发布。每个版本绑定引擎地址和运行时代码哈希。执行前，注册表重新检查引擎字节码和依赖完整性。管理员可以发布新大脑和设置推荐版本，但不能覆盖历史版本，也不能强制所有 AI 迁移。AI 所有者可以手动升级、主动开启自动升级，或永久封印当前大脑。

参考大脑采用有界混合推理：稀疏量化语义路由、显式事实检索、受支持上下文中的 int8 自回归生成，以及对未知问题的确定性回退。模型矩阵和 UTF-8 词表存放在不可变字节码中。每个生成步骤由 EVM 重建激活值、遍历完整词表、选择最高分 token，并把 token 对应的字节拼接为最终回答。

保存对话是 owner-only 状态转换。协议验证 AI 所有权，执行当前大脑，把回答和执行路径压缩为 **traceHash**，然后推进 **turns**、**experience**、**memoryRoot** 和公开记忆环。整个 prompt 和 response 都会进入公开事件，因此协议不提供隐私推理。

组件不是图片属性。它们是有终身供应上限的 ERC-1155 状态变更权。融合会销毁组件，并直接改变目标 AI 的记忆容量、性格、表达范围或技能位。市场则为 AI 和组件提供非托管固定价结算：购买前资产留在卖家钱包；Token Mode 成交时 TINYAI 从买家直接进入卖家钱包，市场本身没有手续费管理入口。

TinyAI 不主张把通用大模型完整塞入 EVM。它提出的是一个更精确的协议原语：一个小型 AI 的执行、身份、状态、升级权和经济交互可以被任何节点复现和验证。

## Abstract

TinyAI Protocol is an EVM-native lifecycle protocol for ownable AI state machines. Each AI is an ERC-721 identity with independent DNA, bounded traits, capability state, a rolling memory commitment, an explicit brain version and an owner-controlled evolution policy.

A brain is not a remote API endpoint. It is an **IBrainEngine** implementation published through an append-only registry. The registry records the engine address and runtime code hash, and revalidates both code identity and locked dependencies before inference.

The reference brain combines sparse quantized semantic routing, bounded retrieval and autoregressive int8 decoding for supported contexts. Model matrices and UTF-8 lexicons are immutable bytecode payloads. Unsupported inputs retain grounded retrieval or an explicit unknown path.

Persistent interaction is an atomic on-chain transition. Only the current AI owner may execute it. The engine returns a response and trace commitment; the protocol then advances experience, turn count, a bounded public memory ring and a rolling memory root.

The result is deliberately narrow but technically falsifiable: training may happen off-chain, while published model data, deterministic inference, final response and persistent state transition execute inside the EVM.

# 1. Protocol definition

TinyAI defines an on-chain AI through five testable properties:

- **Execution sovereignty.** Final response bytes are produced by EVM execution.
- **Model integrity.** The active brain resolves to a fixed runtime identity and locked dependencies.
- **State continuity.** Every saved interaction advances AI-specific public commitments.
- **Owner agency.** The ERC-721 owner controls state writes, migration policy and irreversible sealing.
- **Interface independence.** The protocol remains callable and verifiable without the official website.

A payment record, token-gated interface or server-signed answer is insufficient to satisfy this definition.

## 2. Design principles

### 2.1 The chain is the source of truth

Ownership, DNA, brain version, traits, capabilities, memory commitments and lifecycle policy are protocol state. An interface reads and submits this state but cannot privately redefine it.

### 2.2 Brains are published, not replaced

New engines are appended under sequential versions. Existing entries cannot be overwritten. Runtime code identity is checked at publication and again at use.

### 2.3 Evolution is explicit

Migration is monotonic and controlled by the AI owner. A registry recommendation is not a forced upgrade. Permanent sealing converts an evolvable AI into a version-pinned execution artifact.

### 2.4 Components must change state

A component is valid only when consuming it produces an observable protocol state transition. Decorative metadata does not satisfy the component model.

### 2.5 Settlement is separate from inference

The market transfers protocol assets and accounts for proceeds. It does not control brain execution, AI state or component definitions.

### 3. System architecture

| Layer              | Primitive                        | Responsibility                                                              |
|--------------------|----------------------------------|-----------------------------------------------------------------------------|
| Identity and state | <code>TinyAIProtocol</code>      | ERC-721 ownership, DNA, memory, traits, brain policy and state transitions  |
| Brain publication  | <code>TinyAIBrainRegistry</code> | Append-only versions, runtime commitments and migration eligibility         |
| Inference          | <code>IBrainEngine</code>        | Deterministic routing, retrieval, generation, fallback and trace commitment |
| Model storage      | Bytecode blobs                   | Immutable matrices, dictionaries and UTF-8 lexicons                         |
| Capabilities       | <code>TinyAIComponents</code>    | Capped ERC-1155 effects and protocol-only consumption                       |
| Settlement         | <code>TinyAIMarket</code>        | Non-custodial listings, purchases and seller withdrawals                    |

The primary execution path is:

1. the owner submits `aiId` and prompt to `chatAI`;
2. the protocol validates ownership and prompt bounds;
3. an eligible owner-enabled automatic migration may update the brain version;
4. the registry validates the selected engine;
5. the engine executes against prompt and AI state;
6. the protocol commits the interaction; and
7. the transaction emits complete public execution evidence.

No server callback, oracle request or relayer signature is required.

### 4. AI identity and state

Each AI token maps to an independent state record.

| State                   | Meaning                                      |
|-------------------------|----------------------------------------------|
| <code>dna</code>        | Deterministic identity seed created at mint  |
| <code>memoryRoot</code> | Rolling commitment to persisted interactions |
| <code>bornAt</code>     | Creation timestamp                           |
| <code>experience</code> | Public progression score                     |

| State              | Meaning                                      |
|--------------------|----------------------------------------------|
| brainVersion       | Active registry version                      |
| turns              | Number of persisted conversations            |
| skillMask          | Fused capability bits                        |
| memoryCapacity     | Size of the queryable memory ring            |
| personality traits | Curiosity, empathy, humor and caution        |
| expressionLevel    | Additional deterministic response variants   |
| lifecycle policy   | Automatic migration and irreversible sealing |

DNA is derived from public chain and mint inputs. It differentiates AI state but is not a secret and must not be treated as secure randomness.

The ownership boundary is dynamic. Every state-writing function resolves the current ERC-721 owner, so transferring the token transfers future control without rewriting the state record.

## 5. Deterministic inference

### 5.1 Brain input

The engine receives:

- AI identifier and DNA;
- bounded personality traits;
- expression level and skill mask;
- turn count and memory root;
- speaker address; and
- a prompt of at most 280 bytes.

These fields make behavior conditional on the specific AI while remaining public and reproducible.

### 5.2 Semantic routing

The reference classifier converts bounded UTF-8 input into lexical features, searches a collision-checked sparse dictionary and accumulates quantized intent and sentiment scores.

The router is learned but deterministic. Unknown features are ignored instead of mapped into an uncontrolled collision bucket.

## 5.3 Grounded retrieval

The retriever combines trained semantic scores with explicit entities, cues and negation. It returns bounded fact identifiers only when the evidence threshold is met.

When evidence is insufficient, the engine can choose an explicit unknown path. This is a protocol-level truth boundary, not a frontend message.

## 5.4 Quantized generation

For supported semantic contexts, the decoder uses:

| Parameter                | Bound |
|--------------------------|-------|
| Semantic contexts        | 6     |
| Deterministic variants   | 4     |
| Hidden units             | 32    |
| Vocabulary tokens        | 133   |
| Maximum generated tokens | 16    |

Each decoding step:

1. loads context, previous-token and position embeddings;
2. accumulates a 32-unit activation;
3. scores all 133 vocabulary entries;
4. selects the highest-scoring token;
5. reads its UTF-8 bytes from the immutable lexicon; and
6. repeats until EOS or the token bound.

The complete computation is integer based and deterministic.

## 5.5 Model storage

Model and lexicon payloads are deployed behind an initial **STOP** byte. The payload is inert when called and is read through code-copy operations.

Fixed payload lengths, headers and SHA-256 commitments are validated by the decoder. Engines additionally lock dependency runtime hashes.

## 5.6 Trace commitment

Every inference returns a **traceHash** that commits to the active modules, AI context and selected execution route.

A trace hash is an integrity receipt. It is not private reasoning: all inputs and executed bytecode are public.

## 6. Conversation and memory

**chatAI** is non-payable and owner-only. The protocol exposes no public state-writing room.

For interaction  $n$ :

```
I_n = H(M_(n-1), speaker, H(prompt), H(response), traceHash, blockNumber)
slot_n = (n - 1) mod memoryCapacity
M_n = H(M_(n-1), I_n, slot_n, n)
```

The protocol stores  $I_n$  in the selected ring slot, updates  $M_n$ , advances **turns** and increments experience.

The ring is bounded while the root remains cumulative. Increasing memory capacity extends the number of individually queryable recent interaction hashes without discarding continuity of the rolling root.

An observer can reproduce execution by simulating **chatAI** through **eth\_call** with the current owner as **from** at a fixed block. Simulation does not authorize or persist a transaction.

All real persistent prompts and responses are public. TinyAI must not be used for secrets, credentials or confidential personal information.

## 7. Brain publication and evolution

A brain publication requires:

- deployed engine runtime code;
- the next sequential engine version;
- a bounded label;
- matching engine-reported version; and
- successful engine integrity validation.

The registry stores the engine address, runtime code hash, publication timestamp and migration eligibility.

At execution, the registry rechecks both code identity and integrity. A mismatch fails closed.

AI owners have three policies:

- **Manual migration:** move to a higher enabled version.

- **Automatic migration:** adopt a higher recommendation on the next persistent conversation.
- **Permanent sealing:** pin the current version forever.

The registry owner can publish and recommend. It cannot overwrite a historical entry or force every AI to migrate.

## 8. Component protocol

Components are ERC-1155 assets with immutable on-chain effect definitions.

| Component       | Lifetime cap | State effect                        |
|-----------------|--------------|-------------------------------------|
| Memory Cell     | 30,000       | Memory capacity +1                  |
| Curiosity Gene  | 15,000       | Curiosity +5                        |
| Empathy Gene    | 15,000       | Empathy +5                          |
| Humor Gene      | 15,000       | Humor +5                            |
| Caution Gene    | 15,000       | Caution +5                          |
| Expression Core | 10,000       | Additional deterministic variant +1 |

The catalogue is sealed at a combined lifetime cap of 100,000. Public minting requires the fixed unit price and has no administrative mint path. The six genesis definitions are published at the content-addressed base URI `ipfs://bafybeiduevofpggxaxngsjfzyi4j5uvejtcwkffkh3hgoes6iuomptojm/{id}.json`; changing a later presentation URI cannot change the sealed on-chain effect or lifetime cap.

Fusion requires AI ownership and a non-wasted effect. The protocol burns the component before applying the state change. Burned supply does not reopen historical issuance capacity.

## 9. Market and settlement

**TinyAIMarket** accepts only the designated AI and component contracts.

Listings are non-custodial. Assets remain with sellers until purchase, and ownership, balance and approval are revalidated during settlement.

The Token Mode purchase path is:

1. validate listing, allowance and exact token payment;
2. update remaining quantity or remove the listing;

3. transfer the listed TINYAI amount directly from buyer to seller; and
4. transfer the asset atomically.

ERC-1155 listings support partial fills. ERC-721 listings transfer one AI.

The market has no owner, fee recipient or mutable fee path. It guarantees contract execution rules, not demand or liquidity. The historical Genesis BNB market uses a separate seller **owed** balance; that accounting path is not used by Token Mode.

## 10. Permission model

| Actor           | Permitted actions                                                   | Hard boundary                                              |
|-----------------|---------------------------------------------------------------------|------------------------------------------------------------|
| AI owner        | Persist chat, migrate, configure automatic migration, fuse and seal | Cannot rewrite engines or reverse sealing                  |
| Registry owner  | Publish, recommend and manage new-migration eligibility             | Cannot overwrite versions or force global migration        |
| Protocol owner  | Rotate primary-mint treasury                                        | Cannot change AI cap, mint price or owner-only mutation    |
| Component owner | Initialize catalogue, rotate treasury and update URI                | Cannot admin-mint, change fixed price or exceed sealed cap |
| Market contract | Settle approved protocol assets                                     | Has no owner or fee-setting authority                      |
| Observer        | Read state, events and simulate calls                               | Cannot persist AI state without owner authorization        |

Governance roles use two-step ownership transfer. Direct deployment avoids proxy-slot code replacement, but governance keys remain security-critical.

## 11. Economic surfaces

The deployed Genesis BNB mode fixes:

- maximum AI supply: 10,000;
- AI mint price: 0.0001 BNB;
- component lifetime supply: 100,000;
- component mint price: 0.0001 BNB;
- market protocol fee: zero; and
- conversation protocol fee: zero.

Primary mint proceeds route to configured treasuries. Market purchase value becomes a seller liability. Conversation callers pay network gas directly to validators.

The Flap Token Mode release candidate defines the token as **TinyAI (TINYAI)** and preserves the same 10,000-AI cap, 100,000-component lifetime cap and zero protocol fees for chat and secondary-market settlement, while changing both primary mint prices to exactly 500 TINYAI. It rejects fee-on-transfer underpayment by validating the protocol's exact token balance increase.

The future Flap token is quoted against BSC NVDAB. Buy tax and sell tax are each fixed at 1% for 30 days (2,592,000 seconds) from the launch transaction. The portion of Flap trading tax that reaches the configured beneficiary is routed to a non-custodial AI-holder reward Vault: one currently owned AI represents one equal reward share, rewards do not accrue retroactively to newly minted AI, and unclaimed rewards follow the ERC-721 on transfer. Flap-level deductions occur before beneficiary distribution, so this does not claim that gross trading tax is distributed without deduction.

The deterministic Token Mode Vault, Components, Protocol and Market were deployed and reciprocally bound on BSC mainnet in blocks 117639782-117639985. Their immutable payment-token binding points to **0x30e892840E5E37083c986012934Bd845f8157777**; that address still had no runtime code at preflight block 117640407, and the exact final Flap launch tuple simulated to the same address. Token Mode therefore remains a release candidate until the final Flap token creation and post-launch live-chain checks complete. A Flap salt reservation is an optional front-running mitigation, not a protocol prerequisite: an operator may explicitly choose an unlocked launch, but must accept that a public-mempool observer could copy the salt first. The current BNB Genesis deployment remains the canonical live edition until the final Token Mode evidence is published.

## 12. Security invariants

The implementation is designed around the following invariants:

- AI state mutation requires current ERC-721 ownership.
- Brain versions are sequential and append-only.
- Runtime identity and engine integrity are checked before inference.
- Migration moves only to higher enabled versions.
- Brain sealing is irreversible.
- AI and component caps and prices have no owner setter.
- Component consumption rejects a no-op effect.
- Market state and proceeds accounting update before external asset transfer.
- Token Mode seller proceeds settle directly in TINYAI; the Genesis BNB edition uses pull payments.
- The market has no administrative fee path.
- Persistent interaction evidence is public.

These invariants reduce hidden mutability. They do not prove model quality, protect governance keys, create liquidity or replace independent security review.

## 13. Independent verification

A verifier can work without the official interface:

1. read `ownerOf(aiId)` and `aiState(aiId)`;
2. read the active registry entry;
3. compare the live engine runtime hash with the stored commitment;
4. call `engineFor(version)` to trigger integrity checks;
5. simulate `chatAI` from the owner at a fixed block;
6. repeat the simulation through another compatible node; and
7. for a real transaction, verify `AIChat`, turns, experience, memory root and ring slot.

Illustrative procedure:

```
owner = protocol.ownerOf(aiId)
state = protocol.aiState(aiId)
brain = registry.versionInfo(state.brainVersion)

assert extcodehash(brain.engine) == brain.codeHash

eth_call(
    from      = owner,
    to        = protocol,
    data      = chatAI(aiId, prompt),
    blockTag  = fixedBlock
)
```

Deployment artifacts are useful evidence, but current bytecode, state and receipts are authoritative.

## 14. Composability

TinyAI exposes standard and protocol-specific surfaces:

- ERC-721 ownership and metadata;
- ERC-1155 component balances;
- append-only engine discovery;
- public AI state;
- deterministic inference output;
- event-indexed conversation commitments; and
- protocol-native market settlement.

Other contracts can inspect AI state, gate behavior by ownership or traits, create alternative interfaces, index public memory commitments or build new settlement mechanisms without becoming part of the brain trust boundary.

Composability does not imply unrestricted mutation. Owner-only lifecycle functions remain protected by the ERC-721 authority boundary.

## 15. Research horizon

The current protocol is a bounded execution primitive. Its value is not that it exhausts the design space, but that it establishes an auditable foundation on which more capable systems can be built without dissolving the on-chain trust boundary.

The research horizon includes:

1. **Larger deterministic brains.** Sparse routing, bytecode-sharded parameters, staged retrieval and bounded multi-pass decoding can expand capacity while preserving exact replay at a fixed block.
2. **Hierarchical memory.** Future memory modules may separate short-term context, durable semantic commitments and owner-governed archives while keeping retention and mutation rules independently verifiable.
3. **Proof-carrying inference.** For workloads beyond a single-transaction execution budget, an engine may verify succinct proofs of a committed computation. Such engines must identify proved external computation separately from inference executed natively by the EVM.
4. **Open brain publication.** A standardized engine interface, reproducible model manifests, immutable code commitments and constrained governance can support independently authored brains without allowing silent mutation of existing AI identities.
5. **Contract-native agency.** AI-to-AI messages, bounded asset permissions and capability modules may allow autonomous coordination, provided every authority remains explicit, revocable where promised and limited by ownership policy.
6. **Tokenized resource coordination.** Optional economic modules may allocate inference, storage, curation or governance resources. They are a future extension, not a property of the current zero-fee conversation path.

The north-star is a persistent AI object that can survive the disappearance of any one website, move across compatible applications and adopt better verified brains without losing ownership history or state continuity.

## 16. Conclusion

TinyAI Protocol makes AI a public lifecycle rather than a hidden service.

The NFT supplies identity and ownership. The registry supplies verifiable evolution. The engine supplies deterministic inference. Memory commitments supply continuity. Components supply scarce state transitions. The market supplies native settlement.

The EVM does not need to imitate a data center for this primitive to be meaningful. A small, bounded model can still be genuinely on-chain when its model identity, execution, output and persistent state transition are reproducible by the network itself.

## Appendix A. Core bounds

| Parameter                         | Genesis BNB mode             | Flap Token Mode release candidate |
|-----------------------------------|------------------------------|-----------------------------------|
| Maximum AI supply                 | 10,000                       | 10,000                            |
| Fixed AI mint price               | 0.0001 BNB                   | 500 new Flap tokens               |
| Maximum component lifetime supply | 100,000                      | 100,000                           |
| Fixed component mint price        | 0.0001 BNB                   | 500 new Flap tokens               |
| Maximum prompt size               | 280 bytes                    | 280 bytes                         |
| Maximum memory capacity           | 8 slots                      | 8 slots                           |
| Maximum additional variants       | 2                            | 2                                 |
| Brain migration                   | Higher enabled versions only | Higher enabled versions only      |
| Brain sealing                     | Irreversible                 | Irreversible                      |
| Market fee                        | 0                            | 0                                 |
| Conversation fee                  | 0                            | 0                                 |
| Buy tax                           | Not applicable               | 1% for 30 days                    |
| Sell tax                          | Not applicable               | 1% for 30 days                    |
| Quote / reward asset              | BNB                          | NVDAB                             |

## Appendix B. Terminology

- **Brain:** a registry-published inference engine.
- **DNA:** an AI-specific deterministic identity seed.
- **Memory root:** a rolling commitment to persisted interactions.
- **Memory ring:** a bounded array of recent interaction hashes.
- **Persistent chat:** owner-authorized inference followed by state mutation.
- **Component:** a capped ERC-1155 asset burned to change AI state.
- **Seal:** an irreversible action that pins the active brain.
- **Trace hash:** a compact commitment to execution modules, context and route.
- **Unknown path:** a deterministic response indicating insufficient supported knowledge.